SEFS LAB HPC COMMAND CHEAT SHEET

General command guide for NC State HPC (Hazel) using Slurm scheduler

Cluster Access: ssh <unityid>@login.hpc.ncsu.edu
OS / Scheduler: RHEL 9.2 / Slurm Workload Manager
Author: Michelle Pretorius | **Updated:** Aug 2026

1. FILESYSTEMS & STORAGE DIRECTORIES

Location	Path & Quota	Purpose / Policy
Home	~/ or /home/<unityid> 1 GB QUOTA	Scripts, small apps, dotfiles. Backed up daily.
Scratch	/share/doserlab/<unityid> 20 TB QUOTA	Job execution & raw results. **Not backed up!** Purged after 30 days inactive.
App Dir	/usr/local/usrapps/doserlab/jwdoser	SEFS Lab shared software & R packages repository.
Research	/rs1 or /rsstu 2TB - 30TB	Long-term project data storage across nodes.

2. CONNECTION & INTERACTIVE SFTP

Command	Description
ssh <id>@login.hpc.ncsu.edu	Log in to Hazel via SSH (requires Duo 2FA).
sftp <id>@login.hpc.ncsu.edu	Open interactive secure file transfer session.
exit / Ctrl+D	Safely terminate SSH or SFTP connection.
hostname	Identify current node (loginXX vs compute node).
whoami / clear	Print active UnityID / Clear terminal screen.
Interactive SFTP Sub-commands (inside sftp> prompt):	
put <local> <remote>	Upload file to HPC (add -r for folders).
get <remote> <local>	Download file from HPC (add -r for folders).
lcd <path> / lpwd	Change / print working directory on **local machine**.

3. DIRECTORY & FILE OPERATIONS

Command	Description / Key Options
pwd	Print current full working directory path.
cd <path>	Change directory (cd ~ home, cd .. parent).
ls -ltr	List files in long format sorted by modification time.
mkdir <folder>	Create a new directory (rmdir removes empty dir).
nano <file>	Command-line text editor (Ctrl+X to exit/save).
cat <file>	Print entire file contents to terminal screen.
head -n 20 <f>	Print first 20 lines of a file.
tail -f <f>	Print last 10 lines of a file (-f monitors live).
less <file>	Page through file line-by-line (press q to exit).
grep "str" <f>	Search pattern inside file (e.g. ls | grep txt).
cp -r <s> <d>	Copy file or directory (-r recursive).
mv <s> <d>	Move or rename a file or directory.
rm <file>	Permanently delete file (**Warning: cannot undo!**).

4. QUOTA & MODULE MANAGEMENT

Command	Description
quota -s	Display home directory quota and space used.
quota_display	NC State utility to check group scratch quota on /share .
du -sh .	Summarize total space used by current directory.
module avail	List all software modules available on Hazel.
module load R	Load specific module into active session (e.g. R, gcc).
module list	Show software modules currently loaded.
module purge	Unload **all** modules (best practice at start of scripts).

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5. SLURM BATCH SCRIPT DIRECTIVES (#SBATCH)

Directive Flag	Description / Example
`--job-name=<name>`	Name displayed in queue (e.g. `myjob`).
`--output=stdout.%j`	Standard output log file (`%j` = Job ID).
`--error=stderr.%j`	Standard error log file.
`--ntasks=1`	Number of tasks/processes (MPI ranks).
`--cpus-per-task=4`	Number of CPU cores per task (threads).
`--nodes=1`	Number of compute nodes requested.
`--time=04:00:00`	Wall clock limit (`HH:MM:SS` or `D-HH:MM:SS`).
`--mem=16G`	Memory per node (e.g., `16G` or `16000M`).
`--partition=compute`	Partition selection: `compute` (CPU) or `gpu`.
`--qos=normal`	QOS tier: `normal` (max 4 days) or `long` (max 10 days).
`--array=1-10`	Submit job array with task indices 1 through 10.

6. TEMPLATE SLURM BATCH SCRIPT

```
#!/bin/bash
#SBATCH --job-name=sefs_analysis
#SBATCH --output=stdout.%j
#SBATCH --error=stderr.%j
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --time=02:00:00
#SBATCH --mem=16G
#SBATCH --partition=compute
#SBATCH --qos=normal

# Explicitly switch to submission directory
cd $SLURM_SUBMIT_DIR

# Clean environment & load modules
module purge
module load R

# Execute R script
Rscript ~/scripts/my_model.R
```

7. SLURM JOB MANAGEMENT COMMANDS

Command	Description
`sbatch submit.sh`	Submit batch script to Slurm execution queue.
`salloc`	Start an interactive session on a compute node.
`srun <exec>`	Launch parallel job step (MPI) or command.
`squeue -u $USER`	View status (`ST`) of your queued/running jobs.
`sjs <JOBID>`	View live CPU %, RAM, and disk I/O for running job.
`scancel <JOBID>`	Cancel running job (`scancel -u $USER` cancels all).
`sacct -j <JOBID>`	View accounting details/exit state for completed job.
`seff <JOBID>`	Display CPU and memory efficiency utilization report.
`sa`	Show user account allocations and allowed QOS.
`sqos`	View QOS policies, priorities, and max wall time.
`si`	Show compute node availability across partitions.

8. SLURM STATES & ENVIRONMENT VARIABLES

Code	State	Meaning / Explanation
`PD`	PENDING	Waiting for requested cluster resources or QOS slot.
`R`	RUNNING	Job is actively executing on allocated compute node(s).
`CD`	COMPLETED	Job finished successfully with exit code 0.
`F` / `TO`	FAILED / TIMEOUT	Non-zero exit code or exceeded requested `--time` limit.
`OOM`	OUT_OF_MEMORY	Job process was killed for exceeding `--mem` limit.

Slurm Environment Variables (used inside scripts):

`$SLURM_SUBMIT_DIR`	Directory path where `sbatch` was executed.
`$SLURM_ARRAY_TASK_ID`	Current task index for Job Array (e.g. 1, 2, 3...).
`$SLURM_JOB_ID`	Unique numeric ID assigned to the job.